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Mitochondrial DNA from 3000-year old chickens at the Teouma site, Vanuatu

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ABSTRACT

Chickens were part of the Lapita cultural complex, transported into and through the Pacific by prehistoric colonists; as such they can be used as a proxy for tracking prehistoric migration and interaction. The Lapita site of Teouma in Vanuatu is well known for the recovery of complete dentate stamped pots and a cemetery containing the largest collection of Lapita period skeletons ever found. Chicken bones recovered from these excavations provide the first ancient DNA sequences from any commensal organism directly associated with a Lapita context. The ancient mtDNA sequences obtained from two Teouma chicken bones are compared with previously published archaeologically derived ancient DNA sequences to extend our understanding of the spread of chickens in Pacific prehistory. The results also show that the haplogroup E signature was present in very early populations of chickens transported into Remote Oceania. This study also adds to the suite of available data relating to isotopic signatures for commensal animals during the early settlement of Vanuatu and may reveal a different diet for Teouma chickens than those from other early prehistoric assemblages in the Pacific.

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1. Introduction

1.1. Commensal models and the chicken

The term *commensal* derives from the Latin *commensalis*, *com* meaning together and *mensalis* meaning of the table – interpreted as sharing a table. Fundamentally a commensal model is based on using a phylogeographic approach to document the dispersal patterns of particular species closely associated with the movement of humans, across geographic space (Matisoo-Smith, 1996; Matisoo-Smith and Robins, 2004). All Oceanic domesticates may be considered commensals while not all commensals were domesticated. Consequently both terms may be used to refer to chickens in a Pacific context.

Humans transported a variety of plants and animals across the Pacific in prehistory. The remains of Pacific rats (*Rattus exulans*), pigs (*Sus scrofa*), dogs (*Canis familiaris*) and chickens (*Gallus gallus*) offer unique datasets by which relationships among prehistoric groups of humans may be reconstructed. Using the same approach as has been applied to tracking human migrations (Cann, 2001; Cann et al., 1987), the mitochondrial DNA (mtDNA) of domesticated and commensal species may be used to examine the relationships among ancient populations. Using genetic patterns, historic relationships between populations of plants and animals can be reconstructed. In some cases a geographic signal can also be detected. These historical and geographic relationships can then be extended to understanding the routes by which organisms were transported in prehistory. The approach can be applied to any non-native organism which cannot disperse naturally but which co-occurs in Pacific island archaeological sites with evidence of human activity or settlement. Such organisms are assumed to have been introduced to island environments by humans.

Commensal models, developed initially from studies of the Pacific rat (Matisoo-Smith et al., 1998a), have proven highly

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effective for tracking human migration and interaction in the Pacific. Subsequently the DNA patterns of domesticated animals such as dogs (Allen et al., 1996), pigs (Allen et al., 2001; Larson et al., 2007) and chickens (Storey et al., 2007) have been added. Ancient DNA obtained from commensal animal remains is the preferred unit of study as it allows researchers to construct phylogenetic trees that are temporally relevant. Not only is faunal material generally more abundant in archaeological sites than human bone but the use of animals for the destructive process of ancient DNA analysis is more likely to be supported by local communities (Kirch, 1997; Matisoo-Smith et al., 1998b; Pietruesewsky, 1989).

Ancient DNA data is also critical to studies of the Pacific domesticates as after contact with Europeans a multitude of processes post-dating the archaeological events of interest can hinder accurate reconstructions relying on extant sequence data. In Vanuatu the use of modern chicken mtDNA signatures to reconstruct prehistory is particularly problematic. In 1891 Codrington (1972[1891]), reporting on the preceding decades, remarked that the local native chicken stocks had been fully replaced by European varieties in only 30 years. However, the use of ancient DNA is also complicated and commensal animals must meet several criteria before they can be used in constructing models for prehistoric human behaviour.

In order to be a suitable model organism a commensal species must be unable to disperse naturally between islands, be suitably adaptable to survive in its new environment and be important to the humans who transported it. The chicken is native to Southeast Asia and is not a migratory species (Johnsgard, 1999); it is not capable of sustained flights (Kelly, 2006) or swimming for dispersal between islands. Chickens are found on virtually every continent and are highly adaptable to a range of living conditions (Crawford, 1990). They are used for food, and their feathers and bones have been used for a range of decorative and utilitarian purposes (Steadman, 1997; Steadman et al., 2002).

In Vanuatu, chicken remains have previously been reported from seventeen archaeological sites; including the early sites of Malua Bay (Malakula Island) and Ifo (Erromango) which date to 2860 ± 70 BP (2740–2340 cal BP: ANU-10522, marine shell) and 2700 ± 80 BP (2870–2750 cal BP: ANU-10520, charcoal) respectively (Bedford, 2006; calibrated from the original using OxCal v3.10 (Bronk Ramsey, 2005) with local Vanuatu ΔR of 40 ± 44: Fiona Petchey pers. comm. 2009). Chickens are distributed fairly evenly across the archipelago, but no prehistoric chickens have been reported for Tanna or Aneityum in the extreme south.

1.2. The Teouma site

The site of Teouma on the island of Efate in Vanuatu is on an upraised reef and former beach terrace which has seen five seasons of excavation since its discovery in 2004 (Bedford et al., 2006, 2009). The growing Teouma assemblage includes the largest collection of complete Lapita pots from a single site. The direct association of these pots with a large number of well preserved human burials is also unique (Bedford and Spriggs, 2007; Bedford et al., 2007). The earliest deposits at the site include an extensive Lapita cemetery from which over 70 individuals have been recovered (Bedford et al., 2009; Bentley et al., 2007; authors' data). Extensive studies of this skeletal population have resulted in the identification of dietary constituents (Valentin et al. 2010), have examined issues of local and migrant individuals through isotopic analysis (Bentley et al., 2007) and have provided evidence for gout and other health issues in this prehistoric Pacific population (Buckley, 2007; Buckley et al., 2008).

Archaeological investigations suggest that the initial use of the Teouma site was as a cemetery (Layer 3). However, a small midden located just to the east of the main burial complex was contemporaneous with the burials. The midden contained shellfish and other fauna, the full contents which will be investigated in detail during 2010. The use of the cemetery ceased and in the subsequent Arapus phase the site became used for habitation (Layer 2). Given the flatness of the radiocarbon calibration curve at this time there is almost complete overlap between Lapita and Arapus ^{14}C determinations at this and other Efate sites (Bedford, 2006; Spriggs and Bedford, 2001). A transition date of about 2900 cal BP is postulated on the evidence of a hiatus between cemetery and settlement use, with the cemetery use beginning at about 3100–3000 cal BP (authors' data). The entire settlement was abandoned at some time during the early Erueti phase (2800–2500 cal BP), and the only subsequent use of the site, before the modern period, would have been for gardening (Bedford et al., 2009).

2. Material and methods

Seven chicken bone samples from the site of Teouma were subjected to ancient DNA analyses (Fig. 1, Table 1). Four of the bones came from the 2005 excavation season, one from 2006 and two from 2008. The five bones from the 2005 and 2006 seasons were selected as they had prominent tarsometatarsi, a distinguishing trait of Pacific Island roosters. Two additional elements were recovered by SCH during excavations in 2008 and were brought to



Fig. 1. Photos of the archaeological chicken bone samples from Teouma, Vanuatu which were used in ancient DNA analyses. Provenience details can be found in Table 1.

Table 1
The seven chicken samples from the Teouma site, showing their archaeological provenience, and details of the skeletal element used and its measurements rounded to the nearest millimetre. Elements are illustrated in Fig. 1.

Sample name	Identification number	Site		Layer	Spit/Feature	Element
VUTTEO001	3687	Teouma	Area 2, H6	3	1	Tarsometatarsus
VUTTEO002	3276	Teouma	Area 2, I4	1/2	1	Tarsometatarsus
VUTTEO003	6.914	Teouma	3B.4.6	2	Feature 2	Tarsometatarsus
VUTTEO004	3065	Teouma	Area 2, J2	2	1	Tarsometatarsus
VUTTEO005	3041	Teouma	Area 2, H10	1/2	1	Tarsometatarsus
VUTTEO006	8.2295	Teouma	3A.2.8	3	3 (Burial 50)	Humerus
VUTTEO007	8.652	Teouma	3A.2.9	2/3	3	Sternum

the ancient DNA laboratory directly from the site with no cleaning or processing and as little handling as possible.

2.1. Ancient DNA methodology

All ancient DNA work was conducted in the Ancient DNA facility in the Department of Anthropology at the University of Auckland and undertaken in three physically separated laboratories, one in which ancient DNA was extracted from bone material, a second where Polymerase Chain Reaction (PCR) amplification occurred, and the third where post-PCR purification took place. The methods followed the same protocols detailed in Storey et al. (2007) for a modified guanidine thiocyanate silica suspension technique (Matisoo-Smith et al., 1997). Primers were developed to target a variable portion of the chicken mitochondrial control region and consisted of two pairs which could be used in various combinations:

GG 144F 5′ ACCCATTATATGTATACGGGCATTAA, GG387R 5′ CGAG CATAACCAAATGGGTTAGA, GG218F 5′CATTCACCCTCCCCATAGACAG, GG313R 5′AACCATTCATAGTTAGGAGACTTGTT.

The Ancient DNA Facility at the University of Auckland follows the standards to establish the authenticity of ancient DNA set out by Cooper and Poinar (2000) and Pääbo et al. (2004: 656). Work is conducted in a physically isolated work area, to which access is carefully controlled and all occupants dress in disposable gowns, wear hairnets, face masks, gloves and disposable shoe covers. All stages of work from extraction to PCR amplification include negative controls in which no sample or template aliquots have been added. Ancient DNA template molecules are monitored for appropriate molecular behaviour: we regularly attempt to amplify large fragments (over 350 base pairs (bp)) to ensure there is no modern contamination (Pääbo et al., 2004). Sequences are only considered valid if they are amplified more than once, sequenced in both directions from separate PCR products and the sequences are concordant.

In order to meet authentication standards a sub-sample of VUTTEO003 was also sent to Massey University in Albany (New Zealand) for independent replication of results. At Massey, DNA extraction and amplification were carried out as outlined in Huynen et al. (2003). Amplified products were purified by centrifugation through Sephacryl S200 columns and were cloned into pCR 2.1 (Invitrogen).

2.2. Phylogenetic methodology

The longest sequence from each of the four ancient haplotypes was chosen for alignment using MUSCLE software (Edgar, 2004) with all 169 representatives of the Liu et al. (2006) haplotypes as well as one Ceylonese Junglefowl (EU199948) and two Grey (EU847741 and EU847742) sequences. The resulting alignment was manually checked and trimmed to 175 bp, the length shared by both the ancient and modern datasets.

The overall lack of mtDNA variation in chickens means that there are few phylogenetically informative sites, and phylogenetic trees are expected to have low bootstrap values (Soltis and Soltis, 2003). However, for the sake of comparison with previously published studies on chicken mtDNA, an unrooted Neighbour Joining (NJ) Tree was constructed under the Kimura 2 parameter model with 1000 bootstrap replications using MEGA4 (Tamura et al., 2007). A NJ tree was also constructed under the Kimura 2 parameter model, excluding constant sites using Splits Tree (Huson and Bryant, 2006) and is shown in Fig. 2.

The sequences were also examined through the construction of both bootstrapped and non-bootstrapped Maximum Parsimony (MP) Trees in MEGA4 (Tamura et al., 2007). In the 1000 bootstrap analysis just over 66,200 trees were generated and a random sample of 150 trees was checked for the reconstructed haplogroup affinities of the ancient samples. Finally the MP analysis which did not utilize bootstrapping produced 515 trees all of which were exported to Splits Tree 4 (Huson and Bryant, 2006) to produce a Consensus Network (Holland and Moulton, 2003) using strict MP consensus parameters.

2.3. Radiocarbon dating and isotopes

Sample VUTTEO003 was sent to Rafter Radiocarbon Laboratory and processed by standard methods (cf. Beavan-Athfield and Sparks, 2001). $\delta^{13}C$ and $\delta^{15}N$ stable isotope analysis for gelatine was performed as detailed in Beavan-Athfield et al. (2008). Chicken bone sample VUTTEO006 was sent to Waikato for direct radiocarbon dating. In this instance gelatin was extracted using a modified Longin method (Longin, 1971), the standard pretreatment at the Waikato Radiocarbon Dating Laboratory (Petchey and Higham, 2000). Isotope measurements were undertaken as in Petchey and Green (2005) and as above for the Rafter isotope measurements.

3. Results

As was expected with samples of Lapita age in a tropical environment (Robins et al., 2001) the success rate of DNA amplification from the chicken bones excavated from the Teouma site was low. Of the seven bones from Teouma for which ancient DNA extractions were attempted, two returned reliable, replicable sequences. The

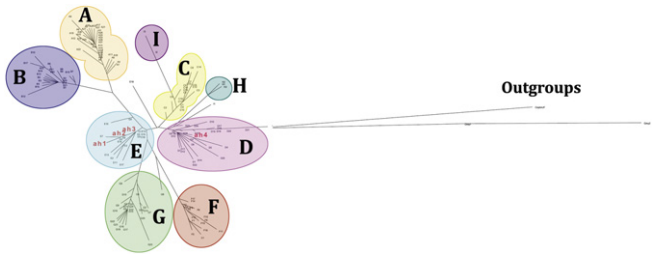


Fig. 2. Neighbour Joining Tree constructed using the ancient data and representatives of each of Liu et al. (2006) haplotypes appended to 175 bp. The ancient haplogroups are highlighted in red. The Ceylonese and Grey Junglefowl were used as outgroups and this branch has been truncated to better fit the page (indicated by the break in the line).

other five did not contain sufficient template material for amplification of mtDNA. The VUTTEO003 bone sample resulted in the independent amplification of an identical at Massey University.

Appropriate molecular behaviour was observed in that only short fragments of mtDNA could be amplified (Pääbo et al., 2004). In total 175 bp were amplified from VUTTEO003 and 201 bp for VUTTEO006. These two sequences were then compared with previously published sequences from Pacific island archaeological contexts (Storey et al., 2007). Given the small sample size and the low number of Single Nucleotide Polymorphisms (SNPs) observed in the sequences, both haplotypes and haplogroups could be assigned using the aligned sequence data. Four distinct haplogroups were observed in the sequence data from archaeological chicken samples. The haplogroups were defined by three distinct SNPs found at sites 213, 278 and 303 (Table 2). Both samples from Vanuatu were of haplogroup I as defined in Storey et al. (2007) but were distinguished from one another at base 321 (NC 001323) where an A to T transversion has taken place distinguishing VUTTEO003 from both VUTTEO006 and the Reference Sequence (Desjardins and Morais, 1990) (see Table 2). Both sequences belong to Liu et al. (2006) haplogroup E and at the truncated length of 175 bp VUTTEO006 is identical to haplotype E6 and VUTTEO003 differs from it by one base (Tables 2 and 3).

In total the chicken sequences, both ancient and modern, were observed to have a low level of variation. Only 69 sites of 175 were variable and of those 43 were parsimony informative. In such cases Woolley et al. (2008) have suggested that with the exception of Minimum Spanning networks most phylogenetic methods are equal in their observable false positive rates (0.05) for the construction of accurate topologies. In examining the results of chicken phylogenies, both the NJ and MP trees placed the ancient samples in the same haplogroups and often grouped them with the same modern haplotypes. Ah1 and ah2 were often affiliated with haplotypes E6 and E7 (Liu et al., 2006) and ah3 was always part of the larger E haplogroup. Haplotype ah4 was consistently affiliated with the D haplogroup. The haplogroup affiliations remained the same in the MP and NJ trees as well as in the construction of a Consensus Network (Fig. 3) (Holland and Moulton, 2003). In each case the affiliations of the modern sequences with haplogroups E and D were as originally described by Liu et al. (2006).

Due to the low diversity of the sequences and their truncated length, the determination of haplogroups was also possible using only a simple alignment of the samples. Through direct observation of alignments of ancient and modern sequences and the dominant SNP locations defined by Liu et al. (2006) ancient haplotypes could easily be assigned to defined haplogroups (See Table 3). The E haplogroup is characterized by a T to C transition at position 214 (numbering based on the Reference sequence NC_001323) and the D haplogroup has an A to G transition which it shares with haplogroup C and a T to C transition shared by all members of the D haplogroup at position 303. Thus even with low statistical support for the phylogenetic reconstructions the examination of the haplogroup defining SNPs confirms the classification of all ancient Pacific chickens with the D and E haplogroups of Liu et al. (2006), contrary to the classification of haplogroup II chickens (Storey et al., 2007) as haplogroup C by Gongora et al. (2008).

The sample VUTTEO003 (which produced the sequence identified as ah1) came from an immediately post-Lapita (layer 2) archaeological feature, a depression or pit filled with dense midden deposits in a dark, organic-rich matrix, dug into the unweathered tephra of Layer 3 in square 3B.4.6. The base of the feature contained a quantity of marine fish bone. The sample VUTTEO006 (which produced a sequence belonging to the ah2 haplotype) was directly associated with Lapita-era Burial 50, and was found within its burial in-filling in square 3A.3.8. Given the early archaeological context of these specimens and recent questions about the validity of haplogroup I (Liu's haplogroup E) as representing an ancient Pacific lineage (Gongora et al., 2008) a portion of each of the samples which resulted in reproducible mtDNA sequences were sent for direct dating.

Sample VUTTEO003 returned a radiocarbon date of 2747 ± 30 BP [NZA 29452]. This was calibrated using OxCal v3.10 (Bronk Ramsey, 2005), using the Northern Hemisphere atmospheric data (Reimer et al., 2004) and resulted in a calibrated two sigma range of 2950–2750 cal BP. The second chicken bone (VUTTEO006) was recovered from a Lapita burial context and was directly dated to 2944 ± 30 BP [Wk 25198], and using the same parameters as above was calibrated to a two sigma range of 3250–2950 cal BP. For ^{14}C purposes the boundary between the atmosphere of the Southern and Northern Hemispheres is considered to lie along the thermal equator, the Inter-Tropical Convergence Zone (ITCZ) and its eastern extension in our region, the South Pacific Convergence Zone (SPCZ) (McCormac et al., 2004:1088). Vanuatu lies just within the SPCZ and so we have used the Northern Hemisphere curve for all Vanuatu terrestrial calibrations.

Isotope measurements were also requested in order to determine if marine contributions to the diet may require the calculation

Table 2
The eleven SNPs identified in the ancient chicken sequences showing the defining sites for each Haplotype. The Haplogroup defining sites of 214, 278 and 303 (Storey unpublished data) are emphasized in bold. Dates for all samples, other than Teouma, are from Storey et al. (2007) (*denotes dates calibrated to 1 σ). When possible the Liu haplotypes with which the ancient sequences were most frequently affiliated is also shown.

Haplotype	Liu et al. (2006) Haplotype	Samples included	Sample age	Acc number	196	214	278	293	303	321	339	443
Ref	A2		Modern	NC 001323	T	**T**	**A**	C	**T**	A	A	C
ah1	E6	VUTTEO003	2950–2750 cal BP	HM189678	C	**C**	•	•	•	T	•	•
ah2	E6, E7	VUTTEO006	3250–2950 cal BP	HM189679	C	**C**	•	•	•	•	•	•
		Tonga TD	2000–1500 cal BP	EF535236								
		HWIKUA001	910–670 cal BP*	EF535238								
		NIUPKI009	1590–1410 cal BP*	EF535239								
ah3	E	PAQANA011	680–550 cal BP	EF535246	•	**C**	•	•	•	•	•	T
		Tonga HB	2000–1500 cal BP	EF535237								
		CHLARA001	750–530 cal BP	EF535241								
ah4	D	PAQANA004		EF535242	•	•	**G**	T	**C**	•	G	T
		PAQANA006	670–550 cal BP	EF535243								
		PAQANA009	660–520 cal BP	EF535244								
		PAQANA010	670–550 and 700–540 cal BP	EF535245								
		PAQHAN001	Prehistoric	EF535247								

Table 3
This table shows the major Haplogroup Defining SNPs of Liu et al. (2006: Figure S1). All the base position numbers in this table are numbered to match the Reference Sequence (NC_001323) and so to compare them with Liu et al. (2006) each position must have three added to it (i.e. 207 in this table is the same as 210 in Figure S1 of Liu et al., 2006).

	1 6 4	2 0 7	2 0 9	2 1 4	2 2 2	2 2 6	2 3 1	2 3 3	2 3 4	2 3 7	2 3 9	2 4 0	2 4 3	2 5 1	2 5 3	2 5 8	2 7 8	2 9 3	2 9 9	3 0 3	3 0 7	3 1 2	3 1 4	3 4 9	3 5 1	3 5 2	3 6 0	3 6 9	4 1 4	4 4 3	4 4 4
NC_001323	C	T	G	T	T	C	C	T	A	C	G	T	C	T	T	C	A	C	C	T	C	C	A	C	T	T	C	A	C	C	T
As	•	C	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•
Bs	T	C	A	•	C	•	•	•	•	•	•	•	T	•	•	•	•	•	•	•	•	T	•	•	•	•	•	•	•	•	•
Cs	T	C	•	•	C	•	•	•	•	A	C	•	•	C	•	•	G	•	•	•	•	•	•	•	•	•	T	•	•	•	•
Ds	T	C	•	•	C	•	•	•	•	•	•	C	•	•	C	T	G	•	•	C	•	•	•	•	•	•	•	•	•	•	•
ah4	T	C	•	•	C	•	•	•	•	•	•	C	•	•	C	T	G	•	•	C	•	•	•	•	•	•	•	•	•	•	•
Es	T	C	•	C	C	•	•	•	•	•	•	•	•	•	•	T	•	•	•	•	T	•	•	•	•	•	•	•	•	•	•
ah1,2 and 3	T	C	•	C	C	•	•	•	•	•	•	•	•	•	•	T	•	•	•	•	T	•	•	•	•	•	•	•	•	•	•
Fs	T	C	•	•	C	•	T	C	•	•	•	C	•	•	C	T	•	•	•	•	T	T	C	•	•	•	•	•	•	T	•
Gs	T	C	A	•	C	•	•	•	•	•	•	C	•	•	C	T	•	A	T	•	T	T	•	•	•	•	•	•	•	T	•
Hs	T	C	A	•	C	T	•	•	•	•	•	C	•	C	•	A	•	•	•	C	T	•	•	T	C	•	•	T	T	T	C
Is	T	C	A	•	C	•	•	•	G	•	•	C	•	•	C	•	G	•	•	•	T	•	•	•	•	C	T	•	•	•	•

• denotes identity with the Reference Sequence.

of corrections for measured radiocarbon dates. Isotope values for VUTTEO003 included both carbon and nitrogen and can be found in Table 4. The isotope data revealed that while the $\delta^{13}C$ is well within a terrestrial range of −20‰, the $\delta^{15}N$ is enriched at 12‰, suggesting that this particular chicken may have acquired a protein rich terrestrial diet. This terrestrial enrichment, however, would have no impact on the measured radiocarbon date. Sample VUTTEO006 had a slightly more enriched carbon signature measured at −18.5‰ and a nitrogen value of 9.8‰, suggesting that this chicken may have derived some of its diet from C_4 plants and/or from marine protein. As a result the Wk 25198 date for VUTTEO006 was calibrated using an estimated 20% marine contribution to the diet. While this likely over estimates the marine component of the diet suggested by the isotope measurements it ensures that the date is not too early (F. Petchey personal communication 2009). With a marine correction of 31 ± 20 the calibrated radiocarbon date provides a two sigma range of 3170–2770 BP.

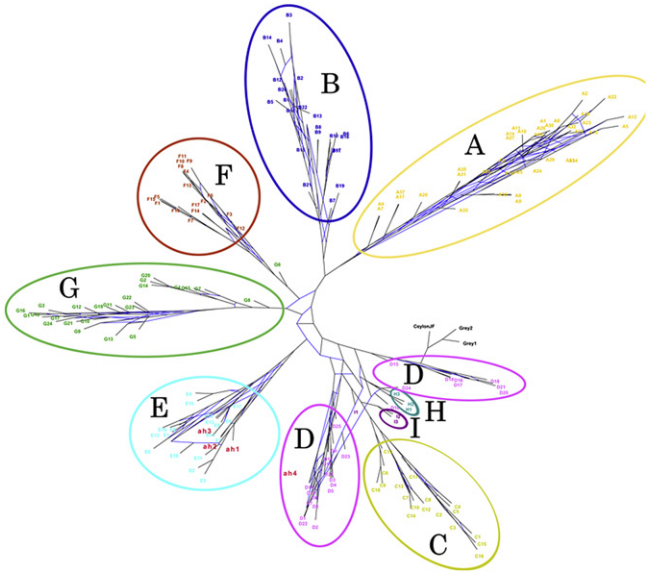


Fig. 3. Consensus Network based on 515 Maximum Parsimony Trees constructed using representatives of each of Liu et al. (2006) haplotypes and each of the ancient haplotypes appended to 175 bp. Ancient haplotypes can be found in Haplogroups D and E and identified by red boxes within the D and E clusters. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

In order to fully assess the marine component of the diet and calculate a more appropriate offset, we attempted to determine a value for the ^{34}S isotope for VUTTEO006. Unfortunately, there was insufficient bone material to allow for the extraction of the 30 mg of bone gelatine required for the accurate measurement of sulphur content. The initial sample, being only a proximal end of a humerus was split into thirds, one for the mtDNA analysis (360 mgs), one (550 mgs/4 mg of gelatine) which generated a radiocarbon date and $\delta^{13}C$ measurement from the Waikato lab and the third (490 mgs/8.5 mg of gelatin), initially submitted for sulphur and used instead to determine values for $\delta^{13}C$ and $\delta^{15}N$ at Rafter Radiocarbon laboratory. Due to issues with the radiocarbon signal for this time period, even with a sulphur determination, any calculated offset may have a significant associated error. Regardless of issues related to fine grained calibrations of the measured radiocarbon date for VUTTEO006, both of the samples are clearly early in age and are archaeologically associated with the Lapita and following Arapus phases of Vanuatu prehistory. This means the mtDNA data can confidently be used in the development of models for the early transport of chickens to Remote Oceania.

3.1. The Teouma mtDNA evidence and prehistoric dispersals

The two mtDNA sequences from Teouma make a valuable contribution to our knowledge of the ancient dispersal of chickens across the Pacific. In the western region of Remote Oceania, haplogroup E is the only lineage that has been detected in archaeological chicken remains. The haplogroup E sequences from Vanuatu are associated with the Lapita cemetery and immediately post-Lapita occupation at the Teouma site. Samples from Tonga, Samoa and Niue are from occupation periods which post-date the Lapita period (Storey et al., 2007). Previously acquired Tongan sequences (Tonga TD and Tonga HB) are from the Plainware period (which begins around 2500 BP and ends at 1550 BP) which followed the Lapita phase (Burley, 1998). The Niuean sample (NIUPKI009) from the Paluki complex dated to 1620 ± 70 BP 1700–1350 cal BP (Walter and Anderson, 2002: [ANU 10738 unidentified charcoal]) is also of haplogroup E. These results, although comprising a small sample size, are all from pre 1000 BP archaeological contexts and thus it is reasonable to surmise that chickens carrying the mtDNA signature for haplogroup E can be affiliated with early Lapita domesticated flocks.

The Near Oceanic source for these birds is difficult to ascertain but Lapita period archaeological deposits in Mussau do contain chicken bones (Steadman and Kirch, 1998; Kirch, 1987). Despite the fact that we have do not as yet know the mtDNA haplogroup to

Table 4
Direct radiocarbon dates and stable isotope values for the Teouma chicken samples which produced reliable reproducible ancient DNA sequences. Radiocarbon ages were calibrated to two σ using OxCal (Bronk Ramsay, 2005) using the Northern Hemisphere Calibration due to the proximity of Efate to the thermal equator.

Sample number	Lab number	Date	Calibrated age 2 σ	wt C%	wt N%	$\delta^{13}C‰$	$\delta^{15}N‰$	C:N
VUTTEO003	NZA 29452	2747 ± 30 BP	2950–2750 cal BP	39.65	13.47	−20.5	12.36	3.4
VUTTEO006	Wk 25198	2944 ± 30 BP	3250–2950 cal BP			−18.5		
				33	11.4	−18.2	9.8	3.4

which the Mussau chickens belong, it would not be unreasonable to hypothesize that members of flocks carrying the haplogroup E signature were moved as part of the Lapita expansion eastward into Remote Oceania, through Vanuatu, beyond to Tonga and Samoa, and then further afield to Niue by Lapita descendants. To date no chicken mtDNA haplogroups other than E have been identified in early, pre 1000 BP contexts in the Pacific, though absence of evidence is not evidence of absence and further analyses of early chicken remains may change our current perception of early chicken dispersals.

Sequences from an early chicken sample from Hawai'i (HWI-KUA001) and another from Easter Island (PAQANA001) (Storey et al., 2007) show that the dispersal of chickens of haplogroup E proceeded eastward across the Pacific, roughly contemporaneously with human expansion to the margins of East Polynesia (Storey et al., 2007). Five of the six sequences derived from chickens recovered from Easter Island archaeological sites were notably different from the haplogroup E sequences amplified from early material from Western Remote Oceania. This second signature, which represents a new haplotype of Liu et al. (2006) haplogroup D most likely, represents a second, temporally distinct introduction of chickens to the Pacific (Fig. 4).

No haplogroup D sequences have yet been observed in materials which predate 1000 BP (Storey, 2009) and since all of the samples from Vanuatu, Tonga, and Niue are from early contexts, it is not clear if the second group of chickens was introduced via the same route as the first. Multiple and temporally distinct introductions have also been suggested for Pacific rats (Matisoo-Smith and Robins, 2004; Tate, 1935), and dogs (Matisoo-Smith, 2007).

The Teouma material firmly establishes a connection between the Lapita cultural complex and haplogroup E chickens. Further work with ancient DNA of commensal organisms from archaeological sites in Vanuatu and other areas of the Pacific will assist in establishing the routes and timing for the introduction of the two ancient haplogroups detected in prehistoric Pacific chickens. While the mtDNA sequences from Teouma do establish a considerable antiquity for haplogroup E in the Pacific and are the only Lapita aged mtDNA sequences for any Pacific commensal, pinpointing the source and origin of this mtDNA signature would be even more informative in the development of a commensal model.

3.1.1. The origins of the Vanuatu chickens

Based on archaeological evidence regarding the dispersal of Lapita-era chickens, the ancient fowl from Teouma are most likely derived from chickens in the Reef and Santa Cruz islands, where other artifacts have indicated contact and interaction between the two areas in prehistory (Green, 1987; Kirch and Yen, 1982; Reepmeyer et al., in press; Spriggs et al., 2010). Chicken bones have been recovered from several Lapita associated archaeological sites in the Reef and Santa Cruz islands including Nendö (Doherty, 2007), Teanu (Kirch, 1983) and the nearby island of Tikopia (Kirch and Yen, 1982). These birds may have come by way of the Bismarck Archipelago where chickens were recovered from two of the Mussau sites (Kirch, 2001) and from Watom, just off New Britain

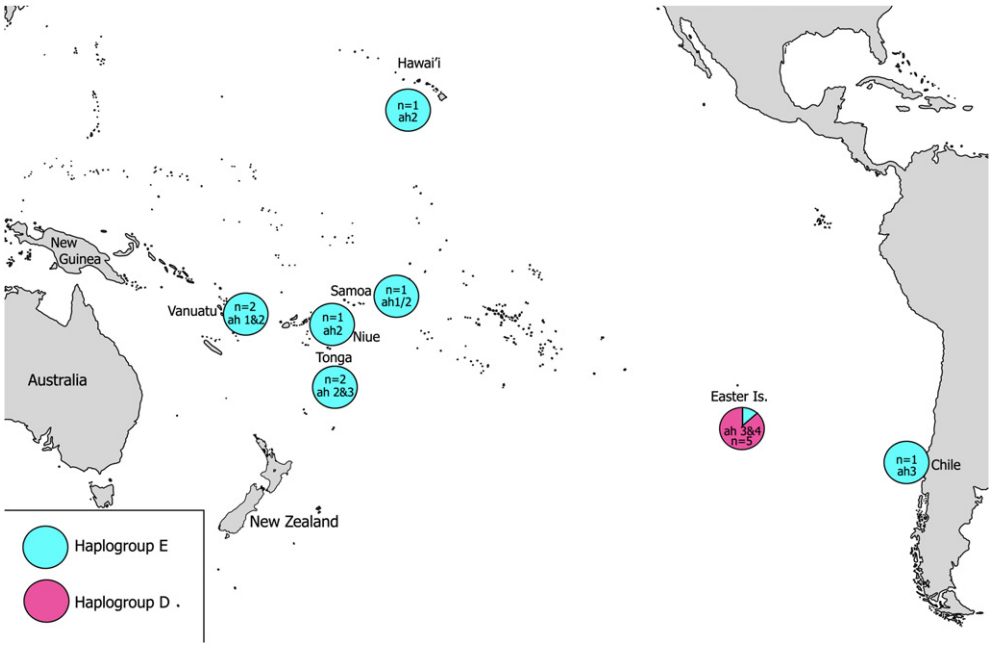


Fig. 4. The distribution of ancient chicken haplogroups across the Pacific. Within each pie chart the information regarding number of samples (n) and ancient haplogroups (ah) represented is noted. The data is not chronologically equivalent with the samples from western Remote Oceania representing individuals which range in age between 3000 and 500 BP and those from Hawai'i, Easter Island and Chile representing samples from post-1000 BP pre-contact archaeological deposits. All data except new sequences from Vanuatu were originally published in Storey et al., (2007).

(Specht, 1968). There is, as yet, no ancient mtDNA evidence or directly dated chicken bones from either of these archipelagos or elsewhere in Near Oceania to connect them with the early Lapita chickens from Teouma. However, the presence of chicken bones from the Reef Island site of Nenumbo (RL-2), one of which has been directly dated to 3047 ± 40 BP (3400–2900 cal BP) (Beavan-Athfield et al., 2008), combined with the aforementioned archaeological evidence for contact, strongly suggest that this archipelago may be the immediate source of the Teouma chickens.

The domestication of chickens likely occurred in multiple locations in Southeast Asia where the Red Junglefowl was native and, since chickens are not indigenous to the Pacific, the origin of Oceanic chickens is undoubtedly Southeast Asian. Therefore the ultimate goal of a commensal model would be to link the Pacific haplogroups back to Asian domestication centres. Such a link might allow for the development of hypotheses about where the chicken component of the Oceanic transported landscape originated. Unfortunately, the archaeological distribution of chicken remains across prehistoric Island Southeast Asia is not well resolved and thus it is not yet possible to trace the transport of chickens from their likely mainland origin, across these archipelagos, and into Near Oceania.

The best sources for genetic information for identifying the probable homelands for domesticated chickens come from the Liu et al. (2006) and Kanginakudru et al. (2008) studies which provide the most comprehensive coverage of modern chicken mtDNA signatures in China and India respectively. However, there are a range of issues which impair the establishment of direct links between the currently available ancient and modern sequence data to develop interpretations about specific origins for ancient chickens in the Pacific. These include the narrow coverage of modern mtDNA data from all of the regions of Southeast Asia to which the Junglefowl is native, a lack of samples from purported domestication centres such as Burma (Crawford, 1990) and issues related to both ancient and modern trade and exchange between probable domestication centres (Hung et al., 2007; O'Reilly, 2007) which is known to have included chickens (Peters, 1913; Qu et al., 2009). The problems are exacerbated by the well known surge in chicken exports from China after 1850 when they were in great demand for the creation of many of the commercial and show breeds which exist today (American Poultry Association, 2001; Stevens, 1991). The fact that all of the chickens included in the Liu et al. (2006) dataset are representative of populations sampled since 1990 means that several events become unresolvable in the phylogeny. These include the primary expansion of chickens from Asian domestication centres in prehistory, secondary movements with the Roman expansion and more recent exports from Asia for intensive breeding programs in the 19th century. None of these are readily distinguishable in the analysis of modern mtDNA signatures. Thus while the origins of ancient Remote Oceanic chickens can be said to be Asian; no more specific homeland can yet be identified with certainty.

Regardless, this study lends further support to the proposition that the ancient sequences from the Pacific, including Vanuatu, are members of haplogroup E (Liu et al., 2006) and represent birds imported to Remote Oceania in the earliest phase of its prehistory (Storey et al., 2007, 2008). Haplogroup E includes modern chickens from China, India, the Middle East and several unprovenienced samples from Europe. Arguments have been made that the similarity of the haplogroup E sequences reported in archaeological samples from the Pacific and Chile with those recorded in modern European chickens ($n = 56$) may indicate European introductions or laboratory contamination (Gongora et al., 2008). Rather than suggesting a European origin for the c.3000–2800 year old mtDNA sequences from Teouma the evidence from the prehistoric Pacific suggests that individuals belonging to haplogroup E are members of a single ancestral lineage from which chicken spread both east and west in prehistory. The ancient DNA sequences and associated radiocarbon dates presented here support the hypothesis that the similarity between the genetic signature of some modern European and ancient Pacific chicken populations points to an ancient maternal ancestor shared by both populations. They do not support either of the alternatives proposed by Gongora et al. (2008).

3.2. Insights from isotopes

The primary aim of examining the ancient Teouma chicken samples was to enhance the existing dataset for ancient chicken mtDNA to create models for the dispersal of chickens to the Pacific in prehistory. However, data gathered from isotope determinations provide additional temporally relevant information which can be used to develop more complex reconstructions of prehistoric human behaviour. Not only can isotopes assist with understanding prehistoric dietary regimes but they show great promise in providing an independent means of examining migration and interaction (Shaw et al., 2009, 2010).

Isotopic data from pigs and rats excavated from Arapus and Erueti phase deposits at Teouma have revealed values for these terrestrial mammals which represent a diet comprised predominantly of C_3 plants (Valentin et al. 2010). This is a surprising result given the propensity of modern Pacific Island pig populations to scavenge along the shoreline for food. In addition the isotope values from Lapita-era humans suggest that marine foods comprised a more significant part of early human diets (Valentin et al. 2010). It is unfortunate that to date the only coeval domesticate sample is from the chicken VUTTEO006 which did result in $\delta^{13}C$ and $\delta^{15}N$ values enriched enough to suggest either a small marine component to the diet (20%) or a significant intake of C_4 plants. This may imply a change in the diet of all of the inhabitants of Teouma over time, as represented by the later faunal isotope values for the single chicken VUTTEO006 and the pig and rat remains sampled by Valentin et al. (2010). Or it may imply that access to marine food scraps and the nearby seashore must have been limited in some way in later periods of prehistory. While in the case of pigs penning or other means of preventing them from rooting through the rubbish or scavenging on the reefs are viable explanations for their terrestrial diets, this is less likely to apply to rats. Why the isotopes measured on the rat populations do not appear to reflect the consumption of marine proteins when they were potentially available in trash heaps and along the low tide seashore is something which is worthy of further more intensive study.

This study adds two chicken samples, one from the Lapita period, to the known commensal isotope dataset (Table 5). As was discussed earlier, in order to ensure that there were no issues with the calibration of the radiocarbon date derived from the chicken bone of the Lapita period a small (20%) marine contribution was calculated based on a $\delta^{13}C$ value of −18.5‰ and a $\delta^{15}N$ of 9.8. There is no significant difference in the absorption of blood gasses and their incorporation into bone by chickens and mammals (Jukes, 1971) so the values measured for the Teouma chickens may be directly compared with values for contemporaneous mammalian fauna. From the later Arapus phase one chicken, one pig and one rat (Valentin et al. 2010) may also represent animals with diets which included a small marine component. However, this does not change the interpretation that their access to marine scraps and rubbish as well as reef resources must have been, in some way, limited. In fact, the chicken bone from the later Arapus period has the most terrestrial signature of all of the commensals from the Teouma site with a $\delta^{13}C$ of −20‰. The enriched value of 12‰ for ^{15}N does

Table 5
Bone collagen isotopic data of animals from the Teouma site.

Area	Sample name	Commensal		Skeletal element	$\delta^{13}C$ ‰	$\delta^{15}N$ ‰	Cultural period
Vanuatu	VUTTEO003	Chicken	*Gallus gallus*	Tarsometatarsus	−20	12	Arapus
	VUTTEO006	Chicken	*G. gallus*	Humerus	−18.5	9.8	Lapita
	TF03[a]	Pig	*Sus scrofa*	Femur shaft	−20.1	9.8	Arapus/Erueiti
	TF04[a]	Pig	*S. scrofa*	Humerus shaft	−18.9	8.3	Arapus/Erueiti
	TF05[a]	Rat	*Rattus exulans*	Long bone shaft	−18.9	8.7	Arapus/Erueiti
	TF06[a]	Rat	*R. exulans*	Cortical bone	−19.4	10.4	Arapus/Erueiti
Reef Islands	SE-RF-2[b]	Chicken	*G. gallus*	Humerus	−15.9	12.9	Lapita
Fiji[c]		Chicken	*G. gallus*		−16.1	8.9	Lapita
		Chicken	*G. gallus*		−15.2	10.6	Lapita

[a] values from the study of Valentin et al. (2010).
[b] values from Beavan-Athfield et al. (2008).
[c] values from Jones and Quinn (2009).

suggest that a significant amount of protein was derived from terrestrial sources.

These observations are, of course, preliminary and more work focusing on a larger cohort of commensal bones representing a larger range of ages is necessary to develop robust conclusions. For example, it will be of interest to determine if the changes observed in the isotope values of the two chickens are due to changes in the diets of individual chickens vs. entire flocks, in penned vs. feral chickens, or in male and female chickens. The changes in diet over time may reflect intensification of agricultural production, chicken rearing or even cultural preferences. In addition isotopes from pigs and rats of the Lapita-era will add to reconstructions of commensal diets in the earliest settlement periods and will provide an important comparison with chicken derived datasets. These are interesting issues which have the potential add to our understanding of human/animal interactions in prehistory.

When compared with the results from other studies of isotopes derived from archaeologically associated chicken bones in the Pacific, the Teouma results are strikingly different (Table 5). Modern observations of pigs, dogs and chickens foraging on reefs at low tide on some Pacific islands certainly suggest this behaviour was likely to have extended into prehistory in situations where animals had free-range foraging options. Results from a chicken from the site of Nenumbo in the Reef Islands and two individuals from Lau, Fiji (Table 5) suggests that chickens in these two island groups on either side of Vanuatu had more access to marine resources (Beavan-Athfield et al., 2008; Jones and Quinn, 2009). While some issues may arise in the direct comparison of these datasets due to variations in local geological and environmental conditions, the expected range of dietary input values for the Pacific is certainly suggestive of a more terrestrial-based diet for the chicken from Lapita-era Teouma than from either Nenumbo or Fiji. The isotope results from Teouma likely reveal a difference in husbandry practices, a conclusion also proposed by Valentin et al. (2010). This hypothesis is of interest in modelling human/animal interactions in prehistory.

4. Conclusion

The study of chicken bones from archaeological sites has the potential to add valuable information to the reconstruction of prehistoric life in the Pacific. The chickens of Teouma are archaeologically associated with material culture of both the Lapita occupation and its immediate successor, the Arapus phase in central Vanuatu prehistory. A subset of these bones have been directly dated to the earliest phase of human history in Remote Oceania, revealing that chickens were purposefully introduced to Vanuatu by the earliest human inhabitants. The mtDNA sequences obtained from the two Teouma chicken samples belong to haplogroup E which is consistent with other early chicken sequences from Pacific sites (Storey et al., 2007) and represent the earliest mtDNA sequences from any commensal organism. Haplogroup E can now be directly associated with Lapita introductions and points back to an ancestral Asian node of domesticated chickens. Once contemporaneous materials from Near Oceania, Island Southeast Asia and the Asian mainland are available for ancient DNA analyses, the source and route of introduction for Lapita-era chickens may become clear.

Stable isotope measurements from the single chicken bone, VUTTEO006, may indicate that in the Lapita period the chickens at Teouma had differential access to marine resources. This might suggest that commensals were somehow controlled and/or restricted from access to reef and midden foraging as was demonstrated for later commensals by Valentin et al. (2010). Isotope studies of chickens in neighboring archipelagos may also hint that the people of Teouma were feeding or treating their chickens differently than people in Nenumbo and Fiji. However, this proposition is based on only a few individuals from three specific sites and should be explored more intensively in future to determine the nature of these differences. This is because the biomolecular study of chicken remains has proven to be a useful tool in the reconstruction of ancient lifeways allowing us to document migration and interaction, as well as some aspects of human/animal relationships in prehistory.

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